



Maximum and Minimum Values

Section 4.1

A. Absolute Maximum or Minimum/Extreme Values

A function $f(x)$ has an **Absolute Maximum** at $x = c$ if $f(c) \geq f(x)$ for every point x in the domain.

Similarly, $f(x)$ has an **Absolute Minimum** at $x = c$ if $f(c) \leq f(x)$ for every point x in the domain.

B. Local/Relative Maximum or Minimum Values

A function $f(x)$ has a **Local Maximum** at $x = c$ if $f(c) \geq f(x)$ for every point x that is near c .

A function $f(x)$ has a **Local Minimum** at $x = c$ if $f(c) \leq f(x)$ for every point x that is near c .

C. The Extreme Value Theorem

If $f(x)$ is continuous on a closed interval $[a, b]$, then $f(x)$ attains both a maximum and a minimum value on $[a, b]$.

D. Fermat's Theorem

If $f(x)$ has a local maximum or minimum at c , and $f'(c)$ exists, then $f'(c) = 0$.

E. Critical Number

A **critical number** of a function $f(x)$, is a number c in the domain such that $f'(c) = 0$ or $f'(c)$ DNE (does not exist).

If $f(x)$ has a local maximum or minimum at c , then c is critical number of $f(x)$.

E. Closed Interval Method

To find **Absolute Maximum or Minimum** of a continuous function $f(x)$ on a closed interval $[a, b]$:

1. Find the values of $f(x)$ at the critical numbers of $f(x)$ in (a, b) .
2. Find the values of $f(x)$ at the endpoints a and b of the interval.
3. The largest of the values of step 1 and 2 is the **Absolute Maximum**
4. The smallest of the values of step 1 and 2 is the **Absolute Minimum**

***The Closed Interval Method is used to find all absolute Maximums and absolute Minimums of a function over a closed interval!**

Examples

1. **(video)** The *critical number* of the function $f(x) = 2x^2 + 12x - 3$ is

2. **(video)** The *critical numbers* of the function $f(x) = 3x^3 - 9x^2 + 15$ is

3. (video) The *critical numbers* of the function $f(x) = x^{\frac{4}{5}}(x+3)$ is _____
4. (video) Consider the function $f(x) = 4x^2 - 2x + 10$ on the interval $0 \leq x \leq 6$.
The absolute maximum value of $f(x)$ (on the given interval) is _____
and this occurs at x equals _____
- and the absolute minimum of $f(x)$ (on the given interval) is _____
and this occurs at x equals _____
5. Find the critical numbers for the following functions: $f(x) = 2x^2 + 9x - 2$

6. Find the critical numbers for the following functions: $f(x) = -2x^3 + 33x^2 - 60x + 11$

7. Find the critical numbers for the following functions: $f(x) = x^{\frac{4}{5}} \cdot (x - 5)$

8. Find the critical numbers for the following functions: $f(x) = (6x - 2)e^{-6x}$

9. Consider the function $f(x) = 3x^2 - 6x + 8$, $0 \leq x \leq 10$.

The absolute maximum value of $f(x)$ (on the given interval) is _____
and this occurs at x equals _____

and the absolute minimum of $f(x)$ (on the given interval) is _____
and this occurs at x equals _____

10. Consider the function $f(x) = 2x^3 + 18x^2 - 162x + 9$, $-9 \leq x \leq 4$.

The absolute maximum value of $f(x)$ (on the given interval) is _____
and this occurs at x equals _____

and the absolute minimum of $f(x)$ (on the given interval) is _____
and this occurs at x equals _____

11. Consider the function $f(x) = x + 2\cos x$ on the interval $0 \leq x \leq \pi$. Find the absolute maximum and minimum of the function.

12. Choose the best reason that the function $f(x) = x^{91} + x^{25} + x^7 + 13x + 2$ has neither a local maximum nor a local minimum.
- a. The function $f(x)$ is always positive.
 - b. The derivative $f'(x)$ is always negative.
 - c. The derivative $f'(x)$ is always positive.
 - d. The highest power of x in $f(x)$ is odd.